Testing the hypotheses

Which of our explanations survive the evidence?



Today

By the end of today, you will have

Recorded a verdict for each hypothesis with its evidence, and responded in writing to a red team.

- Weigh supporting and disconfirming evidence for each hypothesis.
- Make arguments checkable with claim, evidence and warrant.
- Record verdicts honestly: supported, weakened or unresolved.
- Respond to challenge without overstating.

Competencies: F3 · F5 · E6 · J4

Session plan

10 min	**Opening problem**
15 min	**Concept: weighing evidence**
15 min	**Demonstration: Madhyapur verdicts**
45 min	**Thursday challenge: red team review**
25 min	**Own case: testing matrix**
10 min	**Debrief and exit task**

Opening problem

- **You have found three sources that support your favourite hypothesis. What should you look for next?**

Hint: not a fourth supporting source.

The hypothesis testing matrix: Madhyapur (illustrative)

Hypothesis	Supporting evidence	Disconfirming evidence	Verdict	Confidence
H1 contract	Audit and case studies link weight-based pay to mixed collection	One city mixes under per-household pay, because vehicles lack compartments	Supported; the rival is partly true	Medium
H2 containers	Bins were given out in some wards	Separation barely rose where bins were given	Weakened	Medium
H3 processing	No dry-waste facility in the city plan	Waste pickers recover some dry waste informally	Partly supported	Medium
H4 enforcement	Fines rarely issued	Where fines were tried, dumping rose	The rival is supported: fines alone backfire	Low to medium

Illustrative verdicts for the stylised case.

Claim, evidence, warrant

C

Claim

What you assert. Madhyapur: payment by weight is a major reason crews mix separated waste.

E

Evidence

What supports it. An audit finding and two city case studies where payment terms changed collection practice.

W

Warrant

Why the evidence supports the claim. Crews respond to what they are paid for; where pay followed separated streams, practice changed.

If the warrant is missing, the reader cannot tell why the evidence matters. Most weak arguments fail here.

Three verdicts, honestly recorded

1

Supported

Independent sources agree and the strongest rival is weaker. Still not proof.

2

Weakened

Disconfirming evidence is credible. Revise or drop the hypothesis, and say so.

3

Unresolved

Evidence is missing or conflicting. State what would resolve it. This is a legitimate finding.

Red team rules

1	2	3	4	5
Diagnosis	**Evidence**	**Assumptions**	**Losers**	**Delivery**
What if your diagnosis is wrong?	What evidence have you ignored?	What are you assuming without saying so?	Who loses if you are right?	What could stop this being implemented?

Criticism is quality control, not a personal attack. The presenting analyst responds in writing.

Responding to challenge

Defensive

- ○ Repeat the claim more loudly
- ○ Dismiss the source
- ○ Add more supporting evidence

Analytical

- ● Concede what the challenge shows
- ● Adjust the claim to the strength of the evidence
- ● Say what would settle the disagreement

Changing your mind in response to evidence is assessed as a strength.

Activity: Red team review

Stress-test each analyst's strongest finding, and practise responding to challenge without overstating.

Scenario

Each participant presents their strongest finding in three minutes: claim, evidence and warrant. A red team from another pod challenges it with the five red team questions. The presenter then has two minutes to respond and must state any change to the claim.

What to do

1. Presenter: claim, evidence and warrant in three minutes.
2. Red team: the five questions, one minute each, no speeches.
3. Presenter: respond in two minutes, conceding where right.
4. Scribe: record each challenge and response in the pod log.
5. Presenter: write the revised claim before leaving.

Time 45 minutes (Thursday challenge)

Groups Two pods pair up and red-team each other

Output A revised claim for each participant, in the pod log

Your case: Worksheet 6.1

Hypothesis testing matrix

1. For each hypothesis, list supporting and disconfirming evidence, with sources.
2. Record a verdict: supported, weakened or unresolved.
3. Rate your confidence and say why.
4. Write your strongest finding as claim, evidence and warrant.
5. Record the red team's challenges and your revised claim.

Hand in

Worksheet 6.1 by the start of Day 15, for Gate 3.

Quality check

Every hypothesis has a verdict.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. Look for the evidence that could prove you wrong, not more that agrees.
2. Make arguments checkable: claim, evidence and warrant.
3. Record verdicts honestly. Unresolved is a legitimate finding.
4. Concede what a challenge shows. It strengthens the analysis.

Exit task Write down the hypothesis whose verdict surprised you most.

Before the next session

Independent work

Revise your diagnosis in the light of the verdicts. Draft the governing message for tomorrow's memo in one sentence. About 50 minutes.

Read

Toulmin, Stephen. The Uses of Argument. Cambridge: Cambridge University Press, 1958. The chapter on the layout of arguments.

Next

Day 15: Synthesis and the diagnostic memo

Tomorrow you turn the evidence into a two-page memo with a governing message, and submit Gate 3.